Item 1: the E(F_{p^4}) ladder, with plain, D_4 and double-large-prime arms

IC next-10 campaign, item 1 (`reports/Index calculus leads next steps.md`, "### 1."). 2026-10-14. The protocol, `PROTOCOL.md`, was frozen before any data.

Verdict: NEGATIVE for a crossover with rho at any reachable size. The two kill criteria split:

- The **plain arm's constant-factor lead is killed** on these curves. Its linear-algebra constant against rho-negation is 1.44, above the 0.9 kill line.
- The **double-large-prime (DLP) arm's asymptotic lead survives**. DLP changes the exponent and not only the constant. Its linear algebra (LA) grows as #E^0.420 against the plain arm's #E^0.495, and its whole cost as #E^0.333. Both are below the 0.45 kill line.
- But DLP is 1.8–4.5× dearer than plain at every measured size, and the gap widens. It overtakes plain only at an extrapolated #E ≈ 2^219.
- The **D_4 arm is the cheapest** arm at every size, 20–28× below plain. Its extrapolated handover with rho-negation is 2^126 (fitted exponent) to 2^137 (theoretical exponent 1/4).

Every one of the 84 IC logarithms verified (21 curves × 4 IC arms), and all 672 rho walks. The null control (`dlp-null`, the DLP code path with F' = F) reproduced `plain` exactly on all 21 curves. The positive control (the real D_4 Macaulay solver) matched the oracle on 6 of 6 targets.

What ran

Two artefacts:

1. **The ladder.** `examples/ic_next10_01_fp4_ladder.rs`, `ladder` subcommand.
   - Sizes: 7 sizes, p ∈ {101, 181, 269, 397, 521, 769, 1033}, that is $\log_2$ #E = 26.6 to 40.1. Three curves per size (seeds 1–3): 21 curves.
   - Each curve has one planted target, the same for every arm.
   - Arms: `plain`, `d4`, `dlp` and `dlp-null`. Rho ran 16 times unfolded (`rho4_folded(None)`) and 16 times negation-folded (`rho4_folded(Negation)`) on every curve.
   - Raw data: `raw/ladder.jsonl`, one row per curve.
   - Fits: `raw/summary.json`, computed in Rust (`analyze` subcommand). Python only plots.
2. **ecbench.** E(F_{p^4}) is now an ecbench curve kind (`fp4_translation`), with a method `ic.k4_gaudry`. The generic `rho.negation` and `rho.plain` run on E(F_{p^4}) through a `CountedGroup` implementation.
   - Session `ECBS1hf49e5346dc2b`: 4 curves (p = 101, 181, 269, 397, seed 1) × 3 planted targets × 5 arms = 60 runs. All 60 verified by the runner.
   - Audit: `ecbench verify --replay 12` gave "audit OK: 60 records, 60 verified, 12 replays reproduced, 0 problems". The receipt is `ecbench/session-20261014-fp4.audit.json`, SHA-256 `7f39ce78c4b9d0c141c8a3c24e9ba5b7ed6e732b341a0ea23df2ba4ccb4d5e8d`.
   - Isolation level L0 (`--cpus none`, busy host). These are operation counts only; no wall-time statement is made.

The unit and the one modelling step

S is GAE / √r, ecbench's unit. One E(F_{p^4}) addition is 97 F_p multiplications; the field measured exactly 97.0 on every curve.

Decompositions come from an exact enumeration oracle:

1. A table of x(±P_i ± P_j ± P_k) is built once per base.
2. Each trial probes it with R − s·P_l.
3. Hits are recovered through the pair table.

The oracle returns the decomposition set the algebraic solvers return. Each trial is charged the algebraic solve's measured per-trial cost:

- **$C_4(S_4) = 1.213 \cdot 10^{12}$ F_p multiplications** (§11.17), for the plain and DLP arms;
- **$C_4(D_4) = 2.54 \cdot 10^{10}$**, for the D_4 arm. This item's positive control re-measured it at $2.543 \cdot 10^{10}$ (6 solves, $2.538–2.548 \cdot 10^{10}$).

Running the solver itself on every trial was out of budget. One S_4 solve takes 25 min and 3.7 GB; one D_4 solve took 34–49 s single-threaded here, and the ladder ran 700,701 oracle trials.

In ecbench the oracle's own table and probe work is recorded as `*_uncharged`. The harness therefore flags every IC record as a lower bound ("diagnostic only"). LA is charged 16 F_p multiplications per multiplication mod r (§11.16).

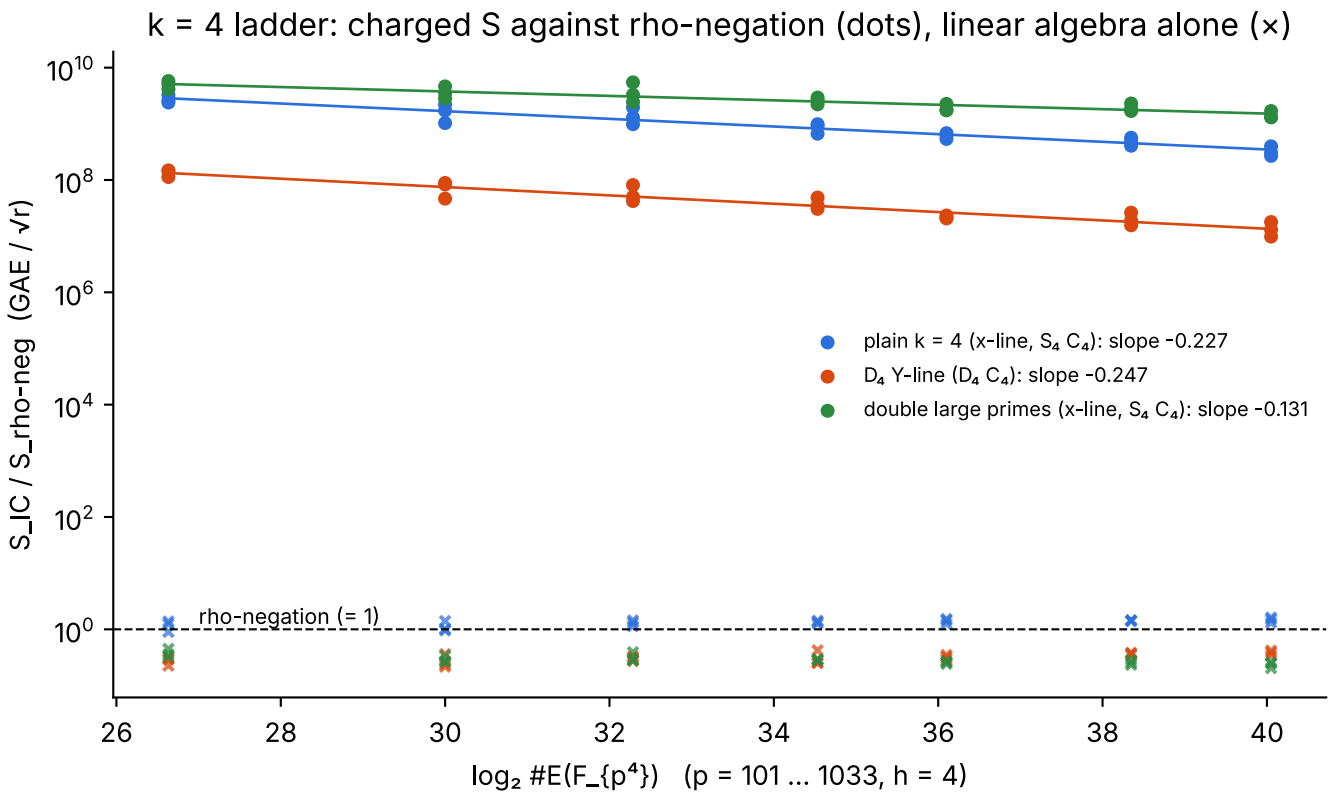
Results

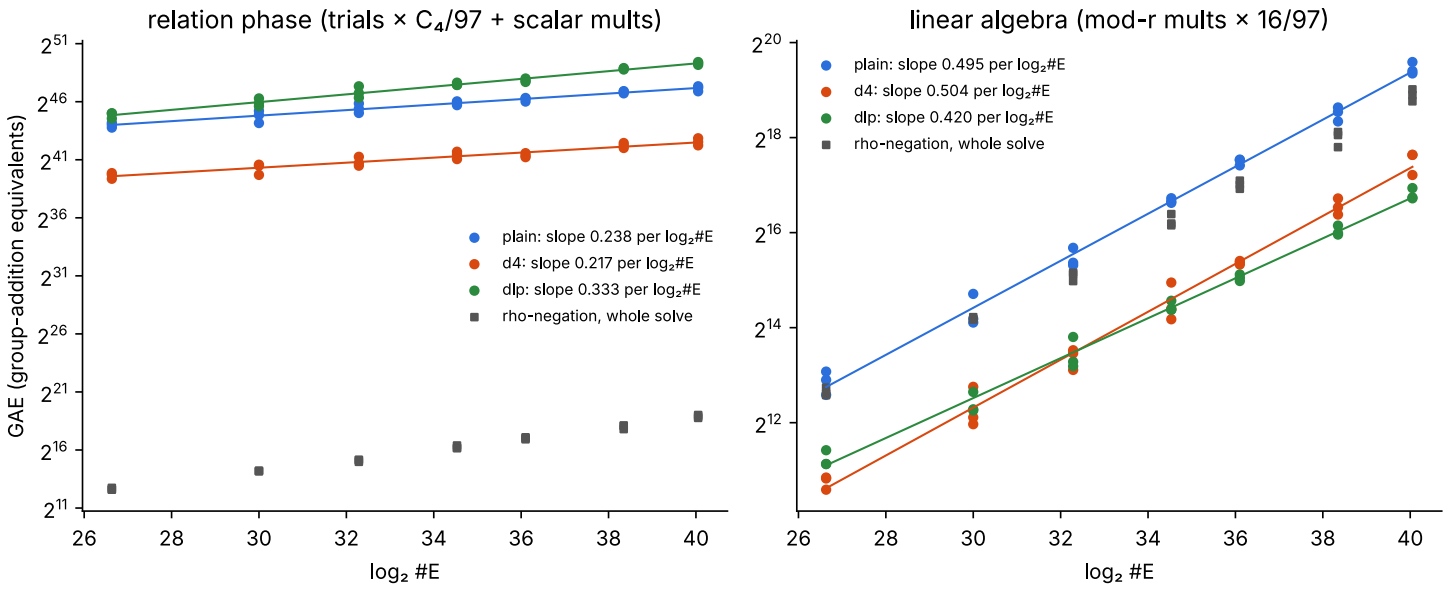
Per size

Means over 3 curves. Measured values: trials, the LA ratio and S. The ratios are derived from them.

p	$\log_2$ #E	$\log_2$ r	S rho	S rho-neg	plain: S / rho-neg	plain: LA / rho-neg	D_4: S / rho-neg	D_4: LA / rho-neg	DLP: S / rho-neg	DLP: LA / rho-neg	plain ÷ D_4	DLP ÷ plain
101	26.6	24.6	1.85	1.26	$2.76 \cdot 10^{9}$	1.17	$1.35 \cdot 10^{8}$	0.27	$5.00 \cdot 10^{9}$	0.38	20.3	1.82
181	30.0	28.0	1.51	1.14	$1.66 \cdot 10^{9}$	1.12	$7.28 \cdot 10^{7}$	0.27	$3.65 \cdot 10^{9}$	0.29	22.7	2.21
269	32.3	30.3	1.29	0.96	$1.41 \cdot 10^{9}$	1.29	$5.78 \cdot 10^{7}$	0.30	$3.73 \cdot 10^{9}$	0.32	24.5	2.64
397	34.5	32.5	1.22	0.99	$8.57 \cdot 10^{8}$	1.34	$3.82 \cdot 10^{7}$	0.31	$2.60 \cdot 10^{9}$	0.29	22.3	3.05
521	36.1	34.1	1.15	0.97	$6.16 \cdot 10^{8}$	1.40	$2.18 \cdot 10^{7}$	0.32	$1.92 \cdot 10^{9}$	0.26	28.2	3.11
769	38.3	36.3	1.25	0.88	$4.87 \cdot 10^{8}$	1.43	$2.03 \cdot 10^{7}$	0.37	$1.95 \cdot 10^{9}$	0.26	24.2	4.01
1033	40.1	38.1	1.31	0.91	$3.24 \cdot 10^{8}$	1.47	$1.36 \cdot 10^{7}$	0.38	$1.44 \cdot 10^{9}$	0.24	24.1	4.45

Columns at p = 1033 were plain 551, D_4 262 and DLP small base 114. Trials at p = 1033 were plain 12,525, D_4 24,830 and DLP 55,718.





Fitted exponents in #E

OLS of $\log_2$(cost) against $\log_2$ #E over 21 curves, with 95% bootstrap intervals over curves.

arm	relation phase	linear algebra	whole charged cost	S / S_rho-neg	LA / rho-neg, pooled at $\log_2$ r ≥ 34
plain	0.238 [0.212, 0.261]	0.495 [0.476, 0.519]	0.238 [0.212, 0.262]	−0.227 [−0.257, −0.198]	1.436
D_4	0.217 [0.191, 0.246]	0.504 [0.483, 0.530]	0.217 [0.192, 0.247]	−0.247 [−0.278, −0.214]	0.357
DLP	0.333 [0.313, 0.353]	0.420 [0.405, 0.440]	0.333 [0.313, 0.353]	−0.131 [−0.157, −0.106]	0.251
rho-negation (whole solve)	—	—	0.464 [0.453, 0.476]	—	—

Theory predicts:

- plain and D_4: relations at #E^{1/4}, LA at #E^{1/2};
- DLP: both phases at #E^{3/8};
- rho: #E^{1/2}.

Rho's measured 0.464 sits below 1/2 because its fixed set-up weighs more at small r.

Against the kill criteria

criterion (report, unchanged)	measured	outcome
plain: slope of S_IC / S_rho shallower than −0.12	−0.227 [−0.257, −0.198]	passes
plain: in-unit LA constant against rho-negation above 0.9	**1.44** at $\log_2$ r ≥ 34; 1.32 ± 0.04 pooled over all 21 curves	**killed** on these curves: plain never crosses rho-negation
DLP: fitted cost exponent in #E ≥ 0.45	whole cost 0.333 [0.313, 0.353]; LA 0.420 [0.405, 0.440]	**passes**: the asymptotic lead stands

The crossover trend (extrapolation, not measurement)

Model:

- rho-negation at its large-size S (0.924 ± 0.080, 9 curves with $\log_2$ r ≥ 34) times √(#E/4);
- each IC relation phase on its own fitted power law;
- the plain and D_4 LA at its pooled large-size ratio, since it is $N^2 \propto \sqrt{\#E}$;
- DLP's LA on its own fitted law.

In the sensitivity column the relation phase is anchored at p = 1033 and carried at the theoretical exponent instead of the fitted one.

arm	IC ≤ rho-negation from (fitted)	anchored at theory exponent	ratio at 2^128	ratio at 2^160	ratio at 2^256
plain	never (LA alone is 1.44× rho-neg)	never	38	1.54	1.44
D_4	**2^125.8**	**2^137.3**	0.77	0.36	0.36
DLP	2^222.5	2^283.3	$5.5 \cdot 10^{4}$	$1.4 \cdot 10^{3}$	0.021
DLP below plain	2^219.4	—	—	—	—

The anchored D_4 handover, 2^137.3, matches `gaudry_quartic_d4_20261013`'s model prediction for h = 4 (2^137.4–2^138.5). This time it comes from measured trial counts rather than the derived rate. The fitted 2^125.8 is lower because the measured trial exponent is 0.217, below 1/4. Its 95% interval, [0.191, 0.246], just excludes 1/4, so the anchored figure is the conservative one.

Does double-large-prime change the exponent?

Yes, and only asymptotically.

- **What moves.** The LA exponent falls from 0.495 (plain) to 0.420. That is above theory's 3/8 because of merge fill-in, as at k = 3 (§11.7: 0.56 against 4/9). The whole-method exponent falls from plain's asymptotic 1/2 to max(0.333, 0.420). Both are below rho's 1/2, so DLP has a real slope win over rho.
- **What it costs.** It buys this with a steeper relation phase: 0.333 against 0.238. At every size measured, and up to the extrapolated 2^219, the relation phase is the entire cost. DLP's LA share at 2^160 is $2 \cdot 10^{-7}$. So at reachable sizes DLP raises the measured exponent and the constant.
- **When it would pay.** It beats plain only once plain's LA dominates (2^219), and beats rho-negation only past 2^222 (fitted) or 2^283 (anchored at 3/8).
- **Ladder versus asymptote.** The exponent change is genuine, but the ladder's measurable consequence runs the other way.

Census (H4): D_4 forces h ≥ 4

`raw/census.jsonl`:

- **What was searched.** 30 instance searches (p = 101, 181, 269, 10 seeds each). Each examined up to about 2,000 candidate curves of the D_4 construction for one with h = 2. The minimum h found was **4 in all 30**. Every one of the 21 ladder curves also has h = 4, as did all 8 curves of E18 (§9.3).
- **Why it is structural.** The construction makes $f'(x_0) = c^2$ with $c \in F_p$. That is the norm, from $L = K(e_2)$ down to K, of $x_0 - e_2$. Over a finite field, the norm $L^*/L^{*2} \to K/K^{*2}$ is an isomorphism. So $x_0 - e_2$ is a square in L, the 2-descent image of T is trivial, and $T \in 2E(K)$. Hence $4 \mid \#E$.
- **What it costs.** The report's "h ≤ 2" is unattainable for the D_4 arm. The cofactor costs √h against rho on every arm run on these curves; plain included, which is why plain's LA constant is 1.44 here. E18 measured 0.733 for plain on prime-order curves.
- **j = 0.** The report also asked about j = 0 with $p \equiv 1 \pmod 3$. $f'(x_0) = 3x_0^2 = c^2$ with $c \in F_p$ forces $x_0 \in F_{p^2}$. Then $b = -x_0^3 \in F_{p^2}$: a subfield curve. So on a non-subfield curve the ψ fold and the D_4 Y-line cannot coexist. This is derived, not measured.

Controls

- **Positive control: the D_4 solver against the oracle.** `raw/d4_control.jsonl`. The real `gaudry_quartic_d4::solve_d4` (Macaulay degree 14) ran on ladder curve p = 269, seed 1, on 3 planted and 3 uniform targets.
  - All 6 closed with quotient dimension 512.
  - Its decomposition sets equal the oracle's on 6 of 6: 8, 8 and 8 on the planted targets, 0 on the uniform ones.
  - $C_4 = 2.538–2.548 \cdot 10^{10}$.
  - The planned second control curve, at the largest p, was not run (time). C_4 does not depend on p.
- **Fit pipeline.** The plain relation phase fits 0.238 [0.212, 0.261] against the expected 1/4.
- **Null control.** `dlp-null` equals `plain` in trials, relations and LA counts on 21 of 21 curves.

ecbench session (in-harness reproduction, 4 sizes)

`ecbench compare`, paired cluster bootstrap over 12 workloads, operations only. The harness labels every IC ratio "diagnostic only (unpriced work)".

B / A	ratio	95% interval
ic-k4-plain / rho-neg	$2.71 \cdot 10^{9}$	[$1.76 \cdot 10^{9}$, $3.92 \cdot 10^{9}$]
ic-k4-d4 / rho-neg	$1.48 \cdot 10^{8}$	[$9.0 \cdot 10^{7}$, $2.17 \cdot 10^{8}$]

B / A	ratio	95% interval
ic-k4-dlp / rho-neg	$6.25 \cdot 10^9$	$[4.34 \cdot 10^9, 8.83 \cdot 10^9]$
ic-k4-d4 / ic-k4-plain	0.0548	[0.048, 0.061]
ic-k4-dlp / ic-k4-plain	2.31	[2.10, 2.62]; per curve 2.01, 2.09, 2.75, 3.18 rising with p
rho-plain / rho-neg	1.92	[1.25, 2.80] (one walk per target)

These agree with the ladder at the same sizes:

- D_4 ÷ plain is about 0.04–0.05 in both;
- DLP ÷ plain rises with p in both.

Full tables are in `ecbench/compare.txt`.

Deviations from the protocol

1. **Cofactor.** The protocol asked for h ≤ 2. Every curve has h = 4, which is structural (the census above).
2. **Stretch size.** p = 1301 was not attempted: the time budget ran out. The largest size is p = 1033, $\log_2 \#E = 40.1$.
3. **Positive control.** It ran on one curve (p = 269), not two.
4. **The LA term in the handover model.** For plain and D_4 the LA term uses the pooled large-size ratio, not its fitted power law. The fitted LA exponents, 0.495 and 0.504, differ from rho's fitted 0.464 only because of rho's set-up at small r. Extrapolating that 0.03 gap over 200 bits would be an artefact. This choice was made after a first look at the data. The first analysis pass used the fitted-law version, which put plain at ratio 0.73 at 2^256.
5. **Degenerate D_4 targets.** These are the 1 in 20 of the D_4 README with quotient dimension 517. They are not modelled: the oracle decomposes them, while the solver would waste the trial. This biases D_4's trial count low by at most about 5%.
6. **Not run.** Plain on prime-order curves (h = 1) was not run here. Its LA constant against rho-negation is E18's 0.733 ± 0.036, which passes the 0.9 line.
7. **Isolation.** The ecbench session ran at L0 with `--allow-busy`. Ten experiments shared 4 cores, with load 17–24. No wall time is used anywhere.

Reproduce

```
        cargo build --release --example ic_next10_01_fp4_ladder --bin ecbench
        ./run_ladder.sh   # 21 curves, RAYON_NUM_THREADS=1, prlimit 2 GB; about 100 min on a loaded 4-core host
        B=target/release/examples/ic_next10_01_fp4_ladder
        $B control --p 269 --seed 1 --planted 3 --uniform 3 --json raw/d4_control.jsonl
        for p in 101 181 269; do $B census --p $p --curves 10 --json raw/census.jsonl; done
        $B analyze --in raw/ladder.jsonl --out raw/summary.json
        python3 -I plot.py
        target/release/ecbench run --wait --allow-busy --cpus none --spec ecbench/spec-fp4-ladder.json --out
ecbench/session-20261014-fp4
        target/release/ecbench verify --dir ecbench/session-20261014-fp4 --replay 12 --out ecbench/session-20261014-
fp4.audit.json
        target/release/ecbench compare --dir ecbench/session-20261014-fp4 --a rho-neg --b ic-k4-d4
```

Source commit 1bb8e17ad plus this item's uncommitted edits (listed below). Host: Claude Code cloud container, 4 vCPU, 15 GiB, shared by ten experiments. Conductor's control plane was unreachable (`conductor check` refused), so there is no task card.

Code touched

- New: `examples/ic_next10_01_fp4_ladder.rs` (ladder, control, census, analyze).
- New: `src/cryptanalysis/ecbench/fp4.rs`, which holds:
  - `Fp4Instance` and its point encoding, four 16-bit limbs per coordinate;
  - `Fp4Group: CountedGroup`;
  - the oracle;
  - `solve_k4`;
  - one unit test.
- Edited, additively:
  - `src/cryptanalysis/ecbench/mod.rs`: `pub mod fp4`.
  - `src/cryptanalysis/ecbench/workload.rs`: `CurveSpec::Fp4Translation` and `Instance::Fp4`, with their match arms. Public targets are refused on fp4.

- `src/cryptanalysis/ecbench/methods.rs`: the `ic.k4_gaudry` registry entry and `K4_PARAMS`, `solve_fp4` dispatch, and the generic rho, BSGS and kangaroo on `Fp4Group`. This changes the IC1 candidate hash, as every `methods.rs` edit does.